INCH-POUND

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PERFORMANCE SPECIFICATION

PROPELLANT, METHANE

This specification is approved for use by all Departments and Agencies of the Department of Defense.



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CONTENTS

PARAGRAPH		PAGE
1	SCOPE	1
1.1	Scope	1
1.2	Classification	1
1.3	Part or identifying number (PIN)	1
2	APPLICABLE DOCUMENTS	1
2.1	General	1
2.2	Government documents	1
2.2.1	Specifications, standards, and handbooks	1
2.3	Non-Government publications	2
2.4	Order of precedence	2
3	REQUIREMENTS	2
3.1	Chemical and physical properties	2
3.2	Qualitative (Type II Only)	2
3.3	Limiting values	2
3.4	Filter	2
3.5	Filled containers (Type I only)	3
3.5.1	Filling pressure	3
3.5.2	Leakage	3
4	VERIFICATION	3
4.1	Classification of inspections	3
4.2	Conformance inspection	3
4.2.1	Examination	4
4.2.2	Sampling tests	4
4.2.3	Individual tests	4
4.3	Sampling plan	4
4.3.1	Sampling size	4
4.3.2	Samplers	4
4.3.3	Sampling methods	4
4.3.4	Cylinders	4
4.3.5	Bulk Conveyance	4
4.4	Rejection and retest	4
4.5	Test methods	4
4.5.1	Examination of product	4
4.5.2	Purity	5
4.5.3	Water	5
4.5.3.1	Electrolytic method	5
4.5.3.2	Piezoelectric method	5
4.5.3.3	Aluminum oxide method	5
4.5.4	Gaseous impurities	5
4.5.4.1	Gas Chromatograph	5
4.5.4.2	Infrared analyzer	5
4.5.4.3	Oxygen analyzer	5
4.5.5	Other gaseous impurities	6
4.5.6	Other volatile hydrocarbons	6
4.5.7	Total volatile sulfur	6
4.5.8	NVR and particulate (Type II only)	6
4.5.8.1	Calculation	6
4.6	Container	7
4.6.1	Filling pressure	7

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4.6.2 Leakage 7

5 PACKAGING 7
5.1 Packaging 7

6 NOTES 8
6.1 Intended use 8
6.2 Acquisition requirements 8
6.3 Packaging requirements 8
6.4 Subject term (key word) listing 8
6.5 Changes from previous issue 8

TABLE
I Chemical and physical properties 3

FIGURE
I NVR apparatus 7

CONCLUDING MATERIAL 9

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1. SCOPE

1.1 Scope. This specification covers two types and three grades of propellant methane (CH_4).

1.2 Classification. The propellant CH_4 will be of the following designated types and grades:

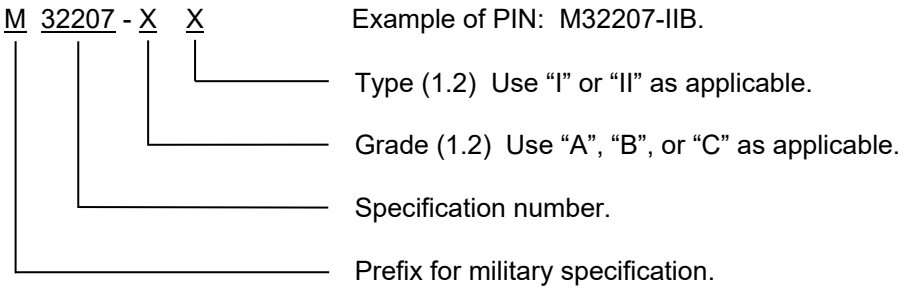
Types

I	Gaseous
II	Liquid

Grades

A	98.7%
B	99.9%
C	99.97%

1.3 Part or identifying number (PIN). The PINs to be used for propellant CH_4 acquired to this specification are created as follows:



2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-27401	Propellant, Pressurizing Agent, Nitrogen
MIL-PRF-27407	Propellant, Pressurizing Agent, Helium

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D1945	Standard Test Method for Analysis of Natural Gas by Gas Chromatography
ASTM D6667	Standard Test Method for Determination of Total Volatile Sulfur in Gaseous Hydrocarbons and Liquefied Petroleum Gases by Ultraviolet Fluorescence
ASTM E29	Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications
ASTM F307	Standard Practice for Sampling Pressurized Gas for Gas Analysis
ASTM F310	Standard Practice for Sampling Cryogenic Aerospace Fluids

(Copies of these documents are available online at https://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Chemical and physical properties. The chemical and physical properties of the propellant shall conform to those listed in Table I when tested in accordance with (IAW) applicable test methods.

3.2 Qualitative (Type II only). The propellant shall be a clear, colorless, homogenous liquid with no solid particles when examined visually by transmitted light at the normal boiling point.

3.3 Limiting values. The following applies to all specified limits in this specification. For purposes of determining conformance with these requirements, an observed value or a calculated value shall be rounded off “to the nearest unit” in the last right-hand digit used in expressing the specification limit according to the Rounding Method of ASTM E29.

3.4 Filter. Unless otherwise specified by the contract or purchase order, a filter with no more than a 10 micrometer (µm) nominal and 40 µm absolute rating shall be installed between the manufacturer’s plant system and the manifold used to fill the gas or liquid containers for delivery.

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TABLE I. *Chemical and physical properties.*

Property	Grade			Test Paragraph
	A	B	C	
Methane (CH_4), % Vol, min	98.7	99.9	99.97	4.5.2
Water, ppmv, max	1	0.5	0.5	4.5.3
Oxygen (O_2), ppmv, max	1	1	1	4.5.4
Nitrogen (N_2), ppmv, max	5000	100	100	4.5.4
Carbon dioxide (CO_2), ppmv, max	125	50	50	4.5.4
Other gaseous impurities, ppmv, max (i.e. Ar, H_2, He, Ne) (1)	5000	125	125	4.5.5
Ethane (C_2H_6), ppmv, max	5000	500	100	4.5.4
Propane (C_3H_8), ppmv, max	3000	500	100	4.5.4
Other volatile hydrocarbons, ppmv, max	1	1	1	4.5.6
Total volatile sulfur, ppmv, max	1	0.1	0.1	4.5.7
Non-volatile residue (NVR) & particulate, mg/L, max (2)	10	1	1	4.5.8
NOTES: (1) Argon (Ar), hydrogen (H_2), helium (He), neon (Ne). (2) Applicable to Type II only.				

3.5 Filled containers (Type I only).

3.5.1 Filling pressure. The filling pressure shall not differ from that required by the contract by more than 1.0 % at 70 °F (21 °C) when tested as specified in 4.6.1. In no case shall it exceed the rated service pressure of the container.

3.5.2 Leakage. Filled cylinders shall not leak when tested according to 4.6.2.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as conformance inspections (4.2).

4.2 Conformance inspection. Conformance tests shall consist of the following:

a. Examination ……4.2.1

b. Sampling tests ……...4.2.2

c. Individual tests …….. 4.2.3

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4.2.1 Examination. Type II propellant shall be sampled according to 4.3 and the sample subjected to tests described under 4.5.

Examination of product …….... ….4.5.1

4.2.2 Sampling tests. The propellant CH_4 shall be sampled according to 4.3 and the samples tested for conformance to the limits of Table I utilizing the procedures described under 4.5. Calibration gas standards may be required to calibrate (zero and span) analytical instruments used to determine the purity and impurity contents of the CH_4. The accuracy of the calibration gas standards is to be traceable to the National Institute of Standards and Technology.

4.2.3 Individual tests. Each Type I container (cylinder, tube or bulk conveyance) shall be subjected to the tests described under 4.6.

4.3 Sampling plan.

4.3.1 Sample size. Each sample shall be of sufficient size (approximately 70 L at STP) to conduct all of the conformance tests as specified herein.

4.3.2 Samplers. The sampler for Type I (gaseous) CH_4 shall be a small compressed gas cylinder. The sampler for Type II (liquid) CH_4 shall be functionally equivalent to a cosmodyne. Liquid samplers convert the entrapped liquid to gas. The aliquots taken for analysis are representative samples.

4.3.3 Sampling methods. Unless otherwise specified, Type I (gaseous) product shall be sampled IAW ASTM F307; Type II (liquid) product shall be sampled IAW ASTM F310. All apparatus used shall be made of suitable materials. Each sample taken for analysis shall be representative of the product being sampled. (NOTE: For Type I (gaseous), a representative sample may also be obtained by withdrawing a sample from the sampling port on the shipping container directly into the analytical instruments).

4.3.4 Cylinders. Unless otherwise specified in the contract or purchase order, each filled cylinder (pressure and Dewar types) offered for shipment shall constitute a lot and be sampled for analysis.

4.3.5 Bulk conveyance. For bulk shipments, each filled container shall constitute a lot. Each individual tube on a tube-bank trailer is considered a separate container. A sample for analysis shall be taken from each portable tank, cargo tank, tank car, or tube filled with Type I (gaseous) or Type II (liquid) methane.

4.4 Rejection and retest. When any sample of the product tested IAW 4.2 fails to conform to the requirements specified herein, the entire lot represented by the sample shall be rejected. Rejected material shall not be resubmitted without furnishing full particulars concerning previous rejection and measures taken to overcome defects.

4.5 Test methods. The samples shall be analyzed according to the procedures described below.

4.5.1 Examination of product. (Type II only) Product shall be visually inspected during the NVR and particulate determination described in 4.5.8 to verify conformance with this specification (3.2).

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4.5.2 Purity. The CH_4 content of the propellant shall be calculated by the following formula:

$$\% \, CH_4 = 100 - \left(\frac{\sum C_1}{10000} \right)$$

where,

$\% \, CH_4$ = purity (% Vol).

$\sum C_1$ = the sum of all impurity species (ppmv).

NOTE: Non-volatile residue (NVR) and particulate are not included in the purity calculation.

4.5.3 Water. The water content of the propellant shall be determined by one of the following methods. In case of dispute, the electrolytic method in 4.5.3.1 shall be the referee.

4.5.3.1 Electrolytic method. Connect the sample container to a pressure regulator which is attached to the electrolytic moisture apparatus (hygrometer). Open the sample container valve and adjust the pressure to the apparatus IAW manufacturer's recommended value. Allow sufficient time for the indicated moisture content to become stable and read the value obtained while using the most sensitive scale setting possible for the moisture content of the sample. The electrolytic moisture apparatus should be set on a range no greater than ten times the specified maximum moisture content.

4.5.3.2 Piezoelectric method. Determine the moisture content utilizing a piezoelectric sorption hygrometer that is set on a range no greater than ten times the specified maximum moisture content by following the instrument manufacturer's instructions.

4.5.3.3 Aluminum oxide method. Determine the moisture content utilizing an aluminum oxide capacitor-equipped analyzer set on a range no greater than ten times the specified maximum moisture content by following the instrument manufacturer's instructions.

4.5.4 Gaseous impurities. Oxygen (O_2), nitrogen (N_2), carbon dioxide (CO_2), ethane (C_2H_4) and propane (C_3H_8) shall be determined by the gas chromatographic method described in 4.5.4.1. The CO_2 content of the propellant may also be determined by the procedure described in 4.5.4.2. The O_2 content may be determined by the procedure described in 4.5.4.3. In case of dispute the gas chromatographic method in 4.5.4.1 shall be the referee.

4.5.4.1 Gas chromatograph. Determine the gaseous impurities of the propellant with a gas chromatograph IAW ASTM D1945. The analyzer shall be capable of separating the compound with a sensitivity of 0.5 ppm or 20% of the specified maximum concentration of the component, whichever is greater. Appropriate impurity concentration techniques may be used to attain the sensitivity. The analyzer shall be calibrated at appropriate intervals by the use of calibration gas standards which contain the applicable limiting characteristic gaseous components in CH_4.

4.5.4.2 Infrared analyzer. The CO_2 content of the propellant may be determined with a gas cell equipped dispersive or non-dispersive infrared analyzer. The analyzer shall be calibrated at appropriate intervals with CO_2 in methane standards at approximately 14.95 µm (669 cm^{-1}). The sensitivity of the analyzer shall be at least 10 ppm or 10 percent of the specified maximum CO_2 content, whichever is greater.

4.5.4.3 Oxygen analyzer. The O_2 content of the propellant may be determined with an electrolytic oxygen analyzer that is set on a range no greater than ten times the specified maximum O_2 content by following the manufacturer's instructions.

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4.5.5 Other gaseous impurities. Calculate the total content of other gaseous impurities of the propellant by summing the amount of argon (Ar), helium (He), hydrogen (H_2), and neon (Ne) as determined utilizing the procedure(s) of 4.5.4.

4.5.6 Other volatile hydrocarbons. The total volatile hydrocarbon content of the propellant, excluding C_2H_4 and C_3H_8 shall be determined utilizing a gas chromatograph with a flame ionization detector.

4.5.7 Total volatile sulfur. The total volatile sulfur content shall be determined IAW ASTM D6667 in which the sulfur compounds are reacted to form a compound which is subsequently measured by UV fluorescence. The analyzer shall be calibrated at appropriate intervals by the use of calibration gas standards. The range of the analyzer shall be no greater than ten times the specified maximum total sulfur content

4.5.8 NVR and particulate (Type II only). Thoroughly clean a 125 mL Erlenmeyer flask. Dry the flask in an oven at 105 ± 5 °C for 30 minutes. After drying, cover the flask with a watch glass or inverted small beaker and allow it to cool next to the balance for 30 minutes. Weigh the flask to the nearest 0.01 mg. Place the flask in a gas purged container that holds the flask securely but allows gas to be purged around the outside of the flask to avoid moisture condensation (Figure 1). Initiate a flow of N_2 gas conforming to MIL-PRF-27401 (Grade B or C) or He conforming to MIL-PRF-27407 (Grade A or B) through the purged container to minimize condensation on the outside of the flask. Fill the flask to the 100 mL line with the liquid CH_4 sample and start a flow of the N_2 or He into the flask to help evaporate the sample and prevent condensation in the flask. Adjust flows as needed to prevent condensation and continue purging until the CH_4 has evaporated and the flask is back to ambient temperature. Shut off gas flows. Remove the flask from the gas purged container, cover with the watch glass or small beaker and place it next to the balance for 30 minutes. Weigh the flask to the nearest 0.01 mg. For Grades B and C, more than 100 mL may need to be evaporated to reduce the effect of balance uncertainty on the result. (NOTE: Sampling should be performed either with the use of a Dewar or directly from the shipping container).

4.5.8.1 Calculation. Calculate the NVR and particulate content of the propellant as follows:

$$NVR\ \&\ particulate\ \ mg/L = \frac{W_g - W_t}{V_s}$$

Where:-

W_g = Gross weight of flask (mg)

W_t = Tare weight of flask (mg)

V_s = Volume of sample (L)

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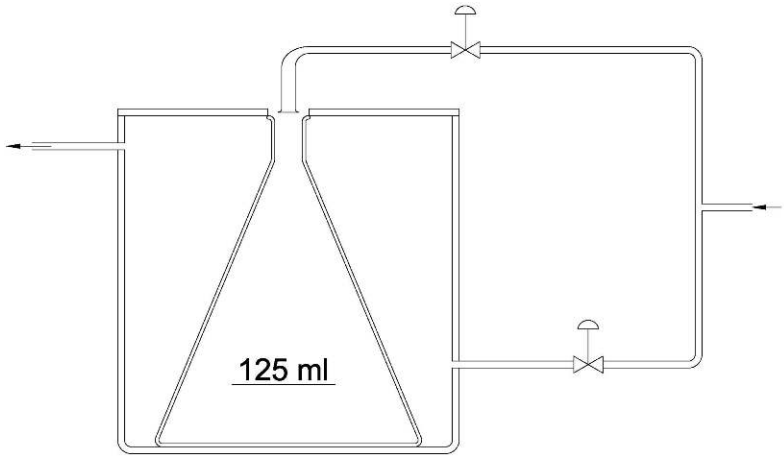


FIGURE 1. *NVR apparatus.*

4.6 Container. For purposes of this paragraph, a container is defined as an individual cylinder, an individual tube in a tube bank trailer, or multiple cylinders that are interconnected by a single manifold that equalizes the pressure across all cylinders.

4.6.1 Filling pressure. Containers shall be tested for proper filling pressure by attaching a calibrated Bourdon-tube gauge or equivalent to the valve outlet and by attaching either a thermocouple or thermometer to the container wall. The gauge shall have scale divisions no greater than 10 psi for service pressures of less than or equal to 3000 psig, and no greater than 15 psi for service pressures greater than 3000 psig. If a thermometer is used, tape or putty shall be applied to the bulb to protect it from extraneous temperatures. Putty shall not be applied between the bulb and the cylinder wall. The thermometer shall have scale divisions no greater than 2 °F (1 °C). The containers shall be stabilized to ambient temperature. Then the valve shall be opened and the internal pressure observed on the gauge.

4.6.2 Leakage. Each container shall be tested for leaks at the neck threads, stem packing, and safety device of the valve with leak-detection fluid. Valve seat leakage shall be tested by means of a tube from the valve outlet to a container of liquid.

5. PACKAGING.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The propellant covered by this specification is intended for use as fuel in rocket engines.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and grade (if applicable) of propellant required (1.2).

d. Method of shipment, type, and capacity of containers.

c. When variation in conformance tests and/or points of inspection are granted (4.2)

d. When variation in sampling method is granted (4.3.3).

6.3 Packaging requirements. Guidance for cylinders may be found in the following documents:

49 CFR 171 - 181	Hazardous Materials Regulations
CGA C-6	Standard for Visual Inspection of Steel Compressed Gas Cylinders
CGA C-6.1	Standard for Visual Inspection of High Pressure Aluminum Alloy Compressed Gas Cylinders
MIL-STD-101	Color Code for Pipelines and for Compressed Gas Cylinders
RR-C-901	Cylinders, Compressed Gas: Seamless Shatterproof, High Pressure DOT 3AA Steel, and 3AL Aluminum

6.4 Subject term (key word) listing.

Cosmodyne

Cylinder

Dewar

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
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Navy – SA
Air Force – 68
DLA - PS

Civil agency:
NASA – NA

Preparing activity:
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